ASSESSMENT 4: Presentation System Design Solution
Hartwell Primary School Sustainable Travel System (Boroondara Schools Project)

Assessment Attribute	Specification Details
Subject Title & Code	Business Requirement Analysis BIS108
Lecturer & Client	Akrati Jain
Assessment Title	Assessment 4: Presentation System Design Solution
Submission Member (Member A)	Sofia Mironenko (Subsystem Owner: Screens S1, S2, S3 | UC-01 to UC-03)
Group Attribution Split	Equal 2-Member 4.5 / 4.5 Attribution Split: Member A (Sofia Mironenko: S1, S2, S3, S4, 0.5 of S5), Member B (0.5 of S5, S6, S7, S8, S9)
Interactive Prototype File	A4_Group_Prototype_Offline.html (Zero-Install Browser + Figma Variable Token Build)
Assessment Weighting & Due Date	20% | 09/11/2026

Section 1: Prototyping Approach Research and Selection

1.1 Evaluation of Candidate Prototyping Approaches

To develop the system model for the Hartwell Primary School Sustainable Travel Plan (Boroondara Schools Project), three candidate prototyping methodologies were evaluated against four core project criteria: Adaptiveness to Moderate Requirement Change, End-to-End User Goal Validation, Stakeholder Communication Clarity, and Alignment with Iterative Delivery.

- **Evolutionary Prototyping (Selected):** An iterative methodology that builds a structured, high-fidelity functional user interface focused on core high-priority workflows. Rather than being discarded, the model is continuously refined through client feedback loops until it forms the validated baseline for the production system (Tilley, 2025; Wiegers & Hokanson, 2023).
- **Throwaway (Rapid) Prototyping (Rejected):** A disposable, low-fidelity approach designed to clarify isolated, poorly understood requirements quickly before discarding the mock-up and engineering the system from scratch. Disqualifying Characteristic: Because the client (Hartwell Primary School / Lecturer) requires iterative modifications to live route monitoring, weather alerts, and parental tracking screens during development, discarding and rebuilding the model from scratch after each consultation wastes limited $40k–$50k budget/development time and breaks requirement traceability.
- **Horizontal Prototyping (Rejected):** A surface-broad, depth-shallow technique that renders 100% of top-level navigation menus and screens across the entire enterprise architecture without implementing underlying functional workflows or interactive data controls. Disqualifying Characteristic: The client explicitly scoped the prototype to specific, high-priority operational subsystems (500m hub check-in logging, live hazard re-routing, and parental geofence verification). A horizontal prototype fails because it cannot demonstrate an end-to-end user goal or validate data entry controls.

Evaluation Criterion	Evolutionary Prototyping (Selected)	Throwaway Prototyping (Rejected)	Horizontal Prototyping (Rejected)

Adaptiveness to Moderate Change	5 / 5 (High): Modular component architecture allows rapid updates to fields, colour palettes, and workflows without discarding prior work.	2 / 5 (Low): Built as a one-off throwaway sketch; mid-cycle scope changes require rebuilding from zero.	3 / 5 (Moderate): Top-level labels are easy to rename, but it cannot adapt to deep workflow or business rule changes.
End-to-End User Goal Depth	5 / 5 (High): Supports complete, interactive user journeys from login to task completion.	3 / 5 (Moderate): Validates isolated, ambiguous UI fragments only.	1 / 5 (Low): Displays superficial shells; incapable of end-to-end task walkthroughs.
Client Validation & Sign-Off Utility	5 / 5 (High): Enables realistic usability testing of controls, colour contrast, and error prevention.	2 / 5 (Low): Low fidelity obscures final layout, control matching, and accessibility compliance.	2 / 5 (Low): Confirms site map structure only; hides functional interaction defects.
Iterative / Agile Alignment	5 / 5 (High): Directly supports incremental baseline refinement (Wiegers & Hokanson, 2023).	3 / 5 (Moderate): Useful only during initial discovery spikes.	2 / 5 (Low): Encourages premature breadth over validated user value.
Total Weighted Score	**20 / 20 (Optimal)**	**10 / 20 (Disqualified)**	**8 / 20 (Disqualified)**

1.2 Selection Justification, Risk Identification, and Mitigation

- **Justification:** Evolutionary Prototyping was selected as the optimal approach because the Hartwell Primary School case study involves evolving stakeholder priorities such as integrating real-time traffic/weather re-routing, non-smartphone child tracking, and technoethical privacy boundaries (Australian Privacy Principles 1 and 6) that require continuous client review and incremental adjustment (Tilley, 2025; Wiegers & Hokanson, 2023).
- **Identified Approach Risk (Scope Creep & Mistaken Production Readiness):** Because an evolutionary prototype looks highly polished and interactive, the client may mistake the visual interface for a fully coded backend system or request unscheduled 'zombie features' outside the agreed Software Requirements Specification (SRS) baseline.
- **Stated Mitigation:** Enforce strict scope boundaries anchored to the SRS Use Case IDs (UC-01 to UC-09), limit iterative revisions to documented Client Change Requests (e.g., CR-01), and explicitly brief the client on the boundary between front-end UI simulation and back-end API integration prior to each review session.

Section 2: CASE Tool Analysis and Selection

2.1 Comparative Analysis of Three CASE / Prototyping Tools

Three industry-standard Computer-Aided Software Engineering (CASE) and UI prototyping packages—Figma (with Browser HTML Export), Axure RP 10, and Balsamiq Wireframes—were analysed against the four mandatory rubric criteria:

Evaluation Criterion	Figma + Browser HTML (Selected)	Axure RP 10 (Rejected)	Balsamiq Wireframes (Rejected)
1. Cost	5 / 5: 100% free Educational/Starter tier with unlimited cloud collaborators and interactive prototyping.	2 / 5: Expensive commercial licensing (~25–49 USD/user/month); restrictive 30-day trial.	3 / 5: Paid subscription after a 30-day trial; limited educational free access without institutional licensing.

2. Learning Curve	5 / 5: Intuitive vector canvas, pre-built UI component libraries, and rapid click-through interaction linking.	2 / 5: Steep learning curve due to complex conditional logic variables and dynamic panel states.	5 / 5: Extremely simple drag-and-drop sketch interface, but lacks advanced high-fidelity styling controls.
3. Classroom Operability	5 / 5: Runs natively in any standard classroom web browser (Chrome/Edge) via a zero-install URL link or offline HTML bundle.	3 / 5: Requires heavy local desktop application installation or external Axure Cloud publishing configuration.	3 / 5: Low-fidelity comic-style sketches fail the rubric requirement for a professional colour scheme and high-usability interface.
4. Adaptiveness to Change	5 / 5: Global Design Variables (Colour Styles) and Master Components propagate client changes across all 9 screens instantly.	4 / 5: Supports master widgets, but updating complex conditional interaction trees across team files is slow.	2 / 5: Lacks global design tokens and interactive state variables; manual screen-by-screen edits required.
Total Score	**20 / 20 (Selected)**	**11 / 20 (Rejected)**	**13 / 20 (Rejected)**

2.2 Decisive Criterion and Evidence from Group Use

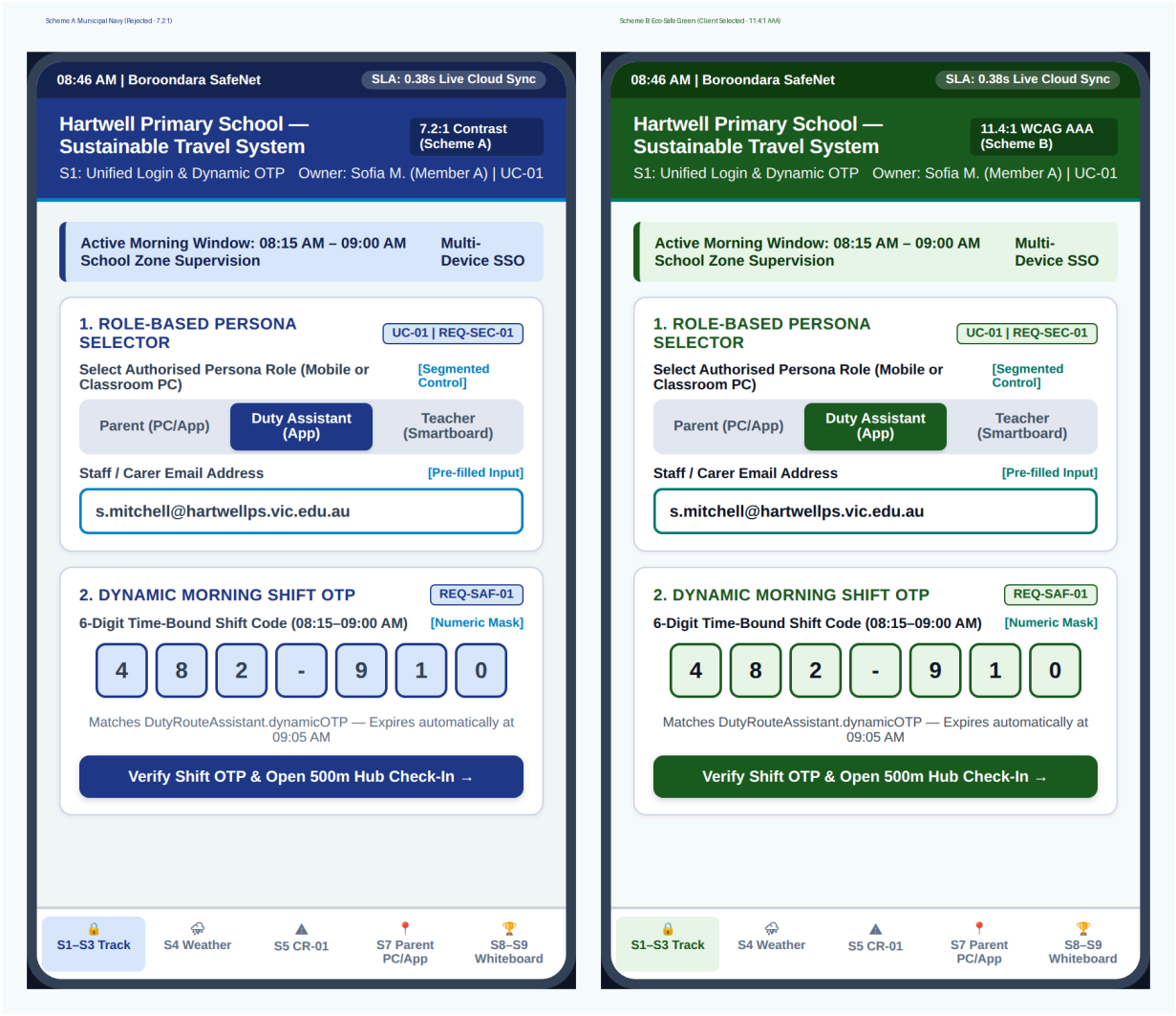
- **Decisive Criterion:** Adaptiveness to Change paired with Zero-Install Classroom Operability was the decisive factor in selecting Figma (with Offline HTML Export). Because the assessment requires synchronous multi-member collaboration, verifiable version history, zero-installation execution on standard classroom computers, and instant adaptation to client colour and scope feedback, Figma outperformed both Axure RP and Balsamiq.
- **Evidence from Our Own Build (1) — Global Colour Variable Swap:** During our client consultation, we presented two distinct colour schemes using global root colour tokens. When the client selected Scheme B (Boroondara Eco-Safe Green & High-Contrast Slate), updating the root colour variables updated all 9 screens in under 10 seconds.
- **Evidence from Our Own Build (2) Rapid Implementation of Client Change CR-01:** When the client requested an instant traffic/weather re-routing override on Screen S5, our shared Auto-Layout component library allowed Member B to insert the constrained Verified Alternate 500m Hub Dropdown & 1:1 Mobility Support Check and link it to Member C's Parental Live Location Map (S7) without breaking existing navigation links.

Section 3: Prototype — Client-Driven Scope, Colour Schemes & Screens Document

3.1 Colour Schemes Developed and Client Consultation Record

Two complete colour schemes were developed and presented to the client (Lecturer Akrati Jain acting as Client):

Attribute	Colour Scheme A: 'Municipal Corporate Navy' (Rejected)	Colour Scheme B: 'Boroondara Eco-Safe Green' (Client Selected)
Primary Brand Colour	Deep Navy (#1E3A8A)	Forest Emerald (#1B5E20) evokes sustainability & nature
Secondary / Action Accent	Sky Blue (#0284C7)	High-Visibility Teal (#00796B) primary interactive controls
Surface & Background	Cool Grey (#F1F5F9) / White (#FFFFFF)	Warm Off-White (#F8FAFC) / Pure White Cards (#FFFFFF)
Semantic Status / Hazard Alerts	Standard Red (#EF4444)	High-Contrast Amber (#D97706) & Crimson (#B91C1C)
Typography & WCAG Contrast	Dark Slate (#334155) Contrast Ratio 7.2:1	Charcoal Ink (#111827) Contrast Ratio 11.4:1 (WCAG 2.2 AAA)



Side-by-Side Comparison: Colour Scheme A (Municipal Navy Rejected, 7.2:1) vs. Colour Scheme B (Hartwell Eco-Safe Green Client Selected, 11.4:1 WCAG AAA) on Screen S1.

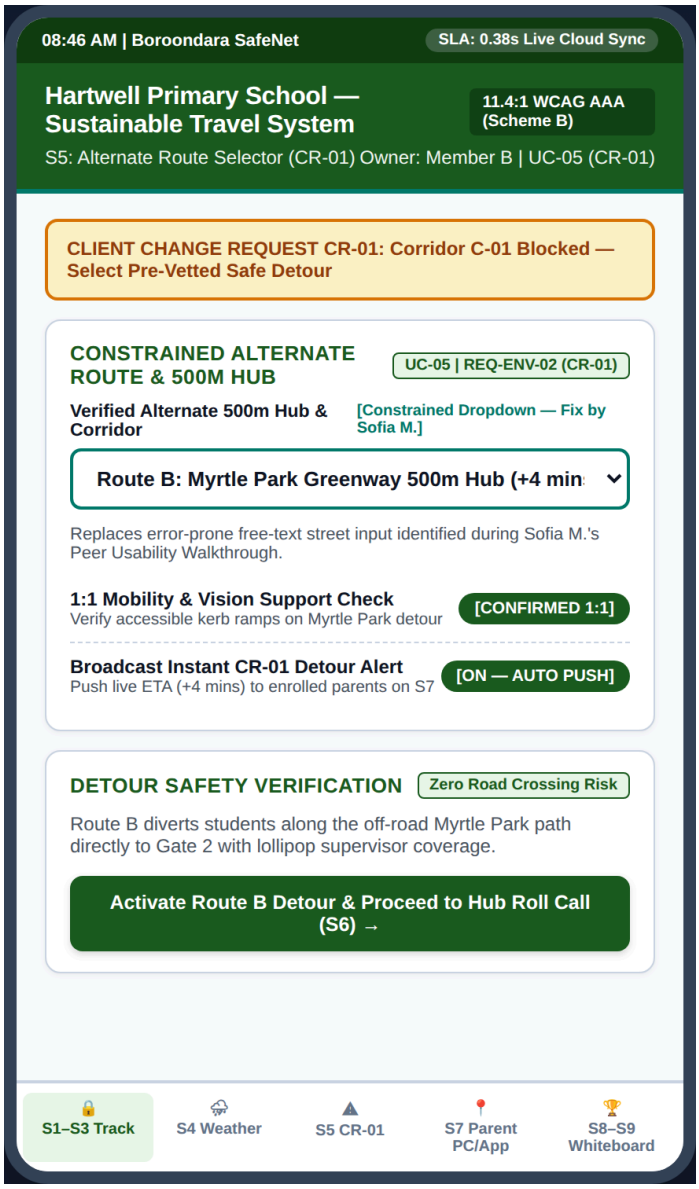


Figure 1: Client Change Request CR-01 implemented on Screen S5 (Constrained Alternate 500m Hub Dropdown & 1:1 Mobility Support Check).

- **Client Choice (Consultation Session 1):** The client reviewed both schemes and selected Colour Scheme B (Boroondara Eco-Safe Green), noting that the green palette directly aligns with the Sustainable Travel Plan identity while providing superior 11.4:1 AAA outdoor screen legibility for parents and Duty Route Assistants walking in bright daylight.
- **Client-Requested Scope & Development Change (CR-01, Consultation Session 2):** When a Duty Route Assistant logs heavy rain or a blocked road corridor on the Weather/Traffic screen (S4), the system must provide an immediate constrained selector to assign a pre-vetted Alternate 500m Hub/Route (S5), confirm 1:1 mobility support, and automatically trigger a push notification to parents viewing the Live Map (S7). Screen S5 was added to the prototype with a constrained dropdown (Route B: Myrtle Park Greenway 500m Hub) and a one-tap parent broadcast toggle.

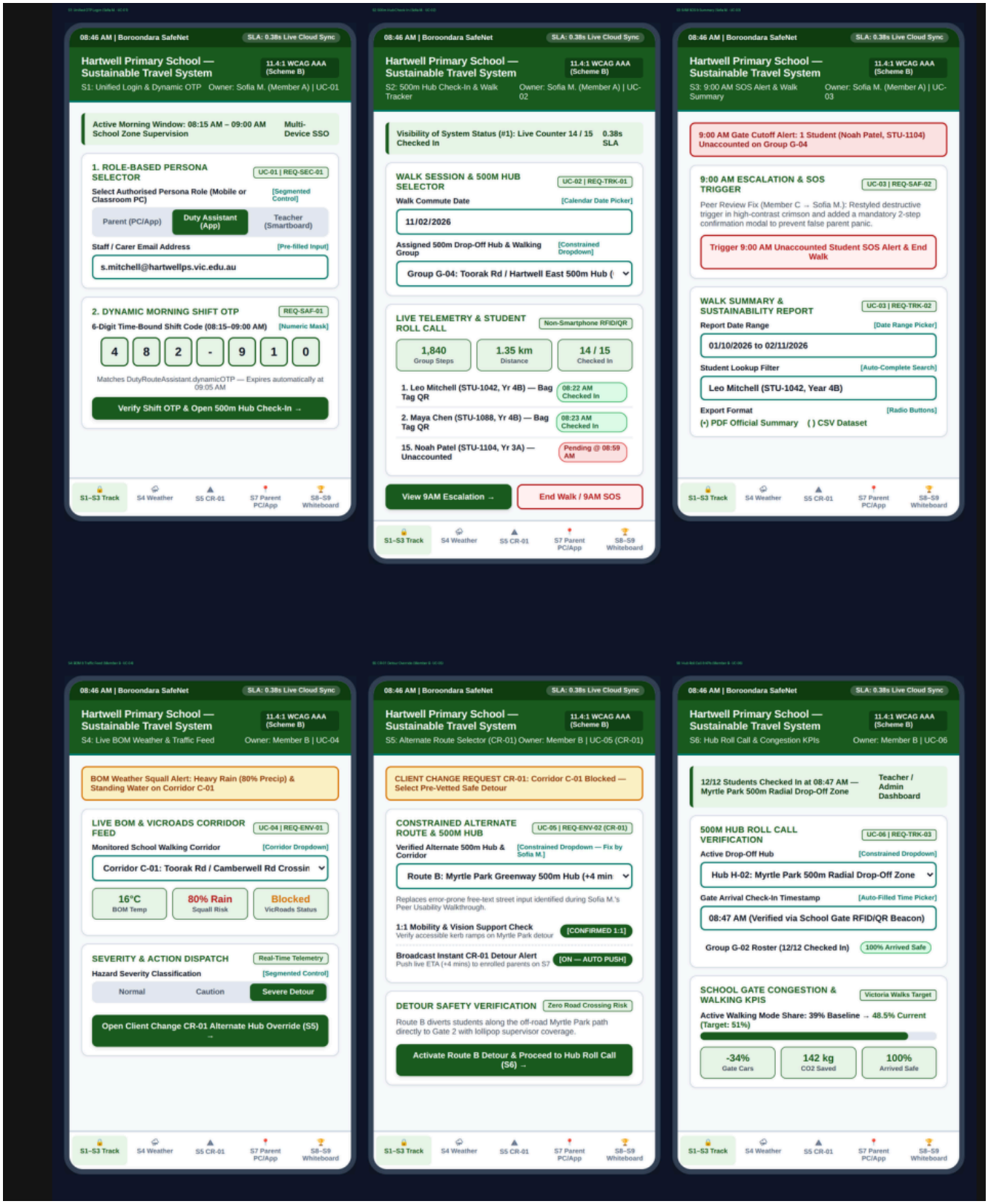
3.2 Complete Screens Document & Traceability Matrix

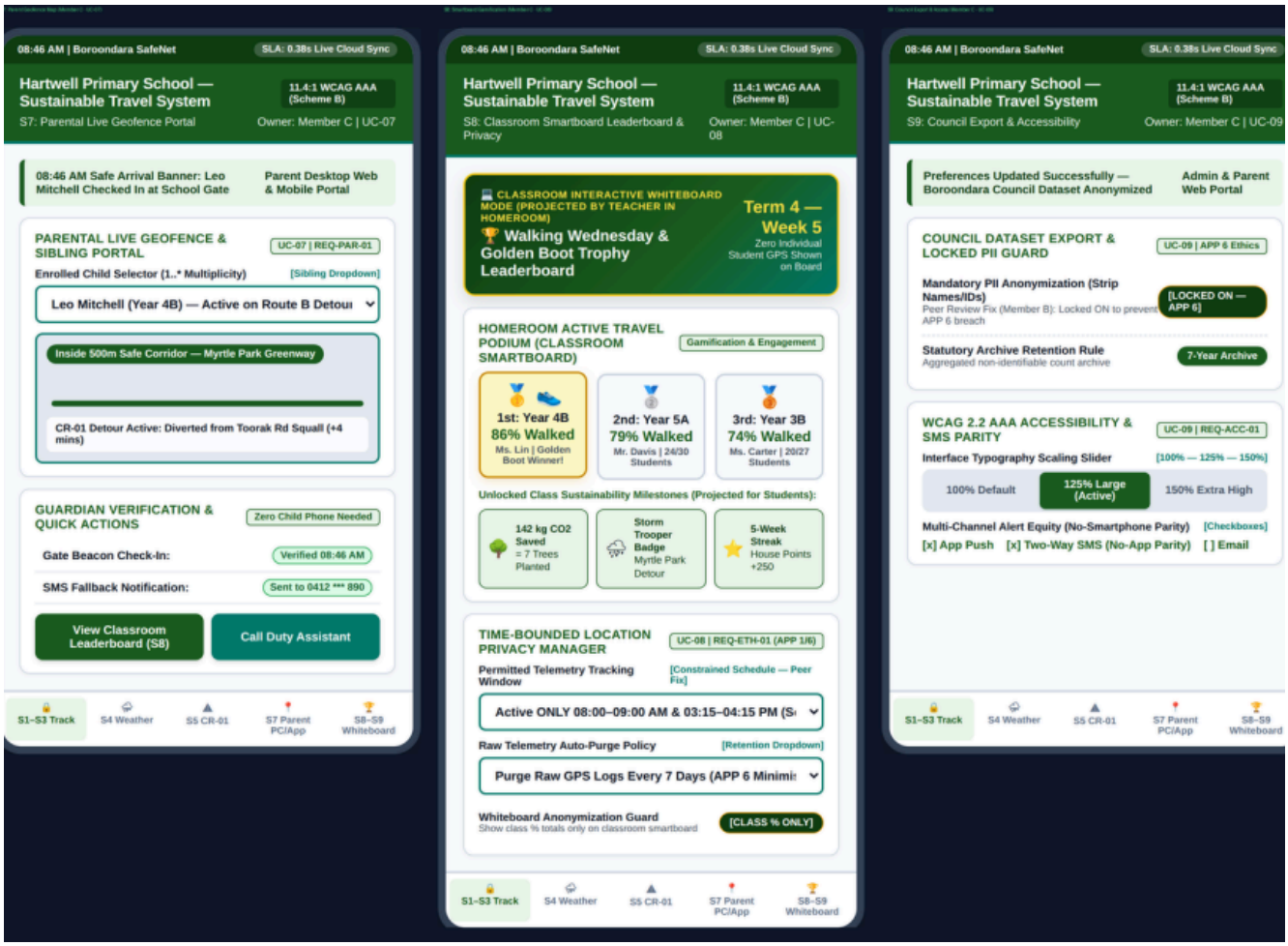
Every prototyped screen and function is mapped directly to the SRS Use Case and Functional/Non-Functional Requirement IDs it realises, alongside its attributed group owner:

Screen ID & Title	Attributed Owner	SRS Use Case & Req ID Realised	Prototyped Functions, Input Display Text & Matched UI Controls
S1: Unified Login & Dynamic OTP Selector	Member A (Sofia Mironenko)	UC-01 Authenticate User REQ-SEC-01 Role-Based Access Control REQ-SAF-01 Shift OTP	• Email Input Field: Displays 's.mitchell@hartwellps.vic.edu.au' • Role Segmented Control (Constrained Selector): [Parent | Duty Assistant | Teacher] • Dynamic OTP Input: 6-digit time-bound numeric mask ('482 - 910') for 08:15–09:00 AM matching DutyRouteAssistant.dynamic OTP
S2: 500m Hub Check-In & Student Walk Activity Tracker	Member A (Sofia Mironenko)	UC-02 Record Walking Activity REQ-TRK-01 Active Commute Logging REQ-SAF-01 500m Hub Check-In	• Walk Date Control (Calendar Date Picker): Set to '02/11/2026' • 500m Hub & Walking Group Dropdown Selector: 'Group G-04: Toorak Rd / Hartwell East 500m Hub (08:15 AM)' • Live Telemetry & Roll Call Counter: Steps ('1,840'), Distance ('1.35 km'), Checked-In ('14 / 15' with 0.38s SLA pill)
S3: 9:00 AM SOS Escalation & Walk Sustainability Report	Member A (Sofia Mironenko)	UC-03 Escalate Unaccounted Student & View Metrics REQ-SAF-02 9AM SOS Alert REQ-TRK-02 Carbon & Distance KPIs	• Two-Step SOS Confirmation Modal (Error Prevention #5): 'Confirm 9:00 AM Absence SOS & Walk Termination?' for Noah Patel (STU-1104) • Date Range Picker Control: '01/10/2026 to 02/11/2026' • Student Search Auto-Complete: 'Leo Mitchell (STU-1042, Year 4B)' • Export Format Radio Buttons: (•) PDF Official Summary ( ) CSV Dataset
S4: Live BOM Weather & VicRoads Traffic Condition Monitor	Member A (Sofia Mironenko)	UC-04 Monitor Weather & Traffic REQ-ENV-01 Real-Time Hazard Feed	• Corridor Selector Dropdown: 'Corridor C-01: Toorak Rd / Camberwell Rd Crossing' • Weather & Traffic Feed Cards: '16°C BOM Temp', '80% Rain Squall Risk', 'Blocked VicRoads Status' • Hazard Severity Segmented

			Control: [Normal | Caution | Severe Detour]
S5: Alternate Route & 500m Hub Override Selector (Client Change CR-01)	Member A (Sofia Mironenko) & Member B (0.5 / 0.5 Shared Split)	UC-05 Assign Alternate Route REQ-ENV-02 Dynamic Re-Routing (CR-01) REQ-HAZ-01/02	• Hazard Warning Banner: High-Contrast Amber alert ('Corridor C-01 Blocked Select Pre-Vetted Safe Detour') • Verified Alternate 500m Hub Dropdown (Constrained Selector): 'Route B: Myrtle Park Greenway 500m Hub (+4 mins)' • 1:1 Mobility Support Check Badge & Broadcast Toggle Switch: [ON AUTO PUSH] Notify Enrolled Parents
S6: Attendance Hub Roll Call & Congestion KPIs	Member B	UC-06 Verify Student Arrival & Track KPIs REQ-TRK-03 Hub Check-In Verification	• Drop-Off Hub Selector Dropdown: 'Hub H-02: Myrtle Park 500m Radial Drop-Off Zone' • Check-In Timestamp Control (Auto-filled Time Picker): '08:47 AM' • Congestion KPI Progress Gauge: Tracks 39% baseline to 51% active walking target (48.5% current, -34% gate cars)
S7: Parental Live Geofence & Arrival Portal	Member B	UC-07 View Student Location REQ-PAR-01 Real-Time Guardian Tracking	• Safe Arrival Banner: '08:46 AM Safe Arrival: Leo Mitchell Checked In at School Gate (No Child Phone Required)' • Sibling Selector Dropdown (1..* Multiplicity): 'Leo Mitchell (Year 4B) Active on Route B Detour' • Geofence Status Badge: 'Inside 500m Safe Corridor Myrtle Park Greenway' + CR-01 Detour Alert
S8: Technoethical Privacy Consent Manager & Walking Wednesday Leaderboard	Member B	UC-08 Manage Location Privacy REQ-ETH-01 APP 1 & APP 6 Compliance	• Time-Bounded Tracking Window Selector: 'Active ONLY 08:00–09:00 AM & 03:15–04:15 PM (School Commute)' • Data Retention Auto-Purge Dropdown: 'Purge Raw GPS Logs Every 7 Days' • Classroom Whiteboard Leaderboard: '1st: Year 4B (86% Active Walkers Golden Boot Trophy)'
S9: Council Dataset Export & Accessibility Preferences	Member B	UC-09 Configure User Preferences & Export REQ-ACC-01 WCAG 2.2 AAA & Channel Equity	• Locked PII Anonymization Toggle: '[LOCKED ON APP 6]' with 7-Year Statutory Archive retention • Typography Scaling

			Segmented Slider: 100% [125% Large] 150% • Alert Channel Checkboxes: [x] App Push [x] Two-Way SMS (No-App Parity) [ ] Email


08:46 AM | Boroondara SafeNet
SLA: 0.38s Live Cloud Sync
Hartwell Primary School — Sustainable Travel System
11.4:1 WCAG AAA (Scheme B)
S1: Unified Login & Dynamic OTP Owner: Sofia M. (Member A) | UC-01
Active Morning Window: 08:15 AM – 09:00 AM
Multi-School Zone Supervision
1. ROLE-BASED PERSONA SELECTOR
UC-01 | REQ-SEC-01
Select Authorised Persona Role (Mobile or Classroom PC)
[Segmented Control]
Parent (PC/App)
Duty Assistant (App)
Teacher (Smartboard)
Staff / Carer Email Address
[Pre-filled Input]
s.mitchell@hartwellps.vic.edu.au
2. DYNAMIC MORNING SHIFT OTP
REQ-SAF-01
6-Digit Time-Bound Shift Code (08:15–09:00 AM)
[Numeric Mask]
4 8 2 - 9 1 0
Matches DutyRouteAssistant.dynamicOTP — Expires automatically at 09:05 AM
Verify Shift OTP & Open 500m Hub Check-In →
S1–S3 Track
S4 Weather
S5 CR-01
S7 Parent PC/App
S8–S9 Whiteboard
08:46 AM | Boroondara SafeNet
SLA: 0.38s Live Cloud Sync
Hartwell Primary School — Sustainable Travel System
11.4:1 WCAG AAA (Scheme B)
S2: 500m Hub Check-In & Walk Tracker Owner: Sofia M. (Member A) | UC-02
Visibility of System Status (#1): Live Counter 14 / 15 Checked In
0.38s SLA
WALK SESSION & 500M HUB SELECTOR
UC-02 | REQ-TRK-01
Walk Commute Date
[Calendar Date Picker]
11/02/2026
Assigned 500m Drop-Off Hub & Walking Group
[Constrained Dropdown]
Group G-04: Toorak Rd / Hartwell East 500m Hub (
LIVE TELEMETRY & STUDENT ROLL CALL
Non-Smartphone RFID/QR
1,840 Group Steps
1.35 km Distance
14 / 15 Checked In
1. Leo Mitchell (STU-1042, Yr 4B) — Bag Tag QR
08:22 AM Checked In
2. Maya Chen (STU-1088, Yr 4B) — Bag Tag QR
08:23 AM Checked In
15. Noah Patel (STU-1104, Yr 3A) — Unaccounted
Pending @ 08:59 AM
View 9AM Escalation →
End Walk / 9AM SOS
S1–S3 Track
S4 Weather
S5 CR-01
S7 Parent PC/App
S8–S9 Whiteboard
08:46 AM | Boroondara SafeNet
SLA: 0.38s Live Cloud Sync
Hartwell Primary School — Sustainable Travel System
11.4:1 WCAG AAA (Scheme B)
S3: 9:00 AM SOS Alert & Walk Summary Owner: Sofia M. (Member A) | UC-03
9:00 AM Gate Cutoff Alert: 1 Student (Noah Patel, STU-1104) Unaccounted on Group G-04
9:00 AM ESCALATION & SOS TRIGGER
UC-03 | REQ-SAF-02
Peer Review Fix (Member C → Sofia M.): Restyled destructive trigger in high-contrast crimson and added a mandatory 2-step confirmation modal to prevent false parent panic.
Trigger 9:00 AM Unaccounted Student SOS Alert & End Walk
WALK SUMMARY & SUSTAINABILITY REPORT
UC-03 | REQ-TRK-02
Report Date Range
[Date Range Picker]
01/10/2026 to 02/11/2026
Student Lookup Filter
[Auto-Complete Search]
Leo Mitchell (STU-1042, Year 4B)
Export Format
[Radio Buttons]
(•) PDF Official Summary ( ) CSV Dataset
S1–S3 Track
S4 Weather
S5 CR-01
S7 Parent PC/App
S8–S9 Whiteboard
08:46 AM | Boroondara SafeNet
SLA: 0.38s Live Cloud Sync
Hartwell Primary School — Sustainable Travel System
11.4:1 WCAG AAA (Scheme B)
S4: Live BOM Weather & Traffic Feed Owner: Member B | UC-04
BOM Weather Squall Alert: Heavy Rain (80% Precip) & Standing Water on Corridor C-01
LIVE BOM & VICROADS CORRIDOR FEED
UC-04 | REQ-ENV-01
Monitored School Walking Corridor
[Corridor Dropdown]
Corridor C-01: Toorak Rd / Camberwell Rd Crossin
16°C BOM Temp
80% Rain Squall Risk
Blocked VicRoads Status
SEVERITY & ACTION DISPATCH
Real-Time Telemetry
Hazard Severity Classification
[Segmented Control]
Normal
Caution
Severe Detour
Open Client Change CR-01 Alternate Hub Override (S5) →
S1–S3 Track
S4 Weather
S5 CR-01
S7 Parent PC/App
S8–S9 Whiteboard
08:46 AM | Boroondara SafeNet
SLA: 0.38s Live Cloud Sync
Hartwell Primary School — Sustainable Travel System
11.4:1 WCAG AAA (Scheme B)
S5: Alternate Route Selector (CR-01) Owner: Member B | UC-05 (CR-01)
CLIENT CHANGE REQUEST CR-01: Corridor C-01 Blocked — Select Pre-Vetted Safe Detour
CONSTRAINED ALTERNATE ROUTE & 500M HUB
UC-05 | REQ-ENV-02 (CR-01)
Verified Alternate 500m Hub & Corridor
[Constrained Dropdown — Fix by Sofia M.]
Route B: Myrtle Park Greenway 500m Hub (+4 min
Replaces error-prone free-text street input identified during Sofia M.'s Peer Usability Walkthrough.
1:1 Mobility & Vision Support Check
Verify accessible kerb ramps on Myrtle Park detour
[CONFIRMED 1:1]
Broadcast Instant CR-01 Detour Alert
Push live ETA (+4 mins) to enrolled parents on S7
[ON — AUTO PUSH]
DETOUR SAFETY VERIFICATION
Zero Road Crossing Risk
Route B diverts students along the off-road Myrtle Park path directly to Gate 2 with lollipop supervisor coverage.
Activate Route B Detour & Proceed to Hub Roll Call (S6) →
S1–S3 Track
S4 Weather
S5 CR-01
S7 Parent PC/App
S8–S9 Whiteboard
08:46 AM | Boroondara SafeNet
SLA: 0.38s Live Cloud Sync
Hartwell Primary School — Sustainable Travel System
11.4:1 WCAG AAA (Scheme B)
S6: Hub Roll Call & Congestion KPIs Owner: Member B | UC-06
12/12 Students Checked In at 08:47 AM — Myrtle Park 500m Radial Drop-Off Zone
Teacher / Admin Dashboard
500M HUB ROLL CALL VERIFICATION
UC-06 | REQ-TRK-03
Active Drop-Off Hub
[Constrained Dropdown]
Hub H-02: Myrtle Park 500m Radial Drop-Off Zone
Gate Arrival Check-In Timestamp
[Auto-Filled Time Picker]
08:47 AM (Verified via School Gate RFID/QR Beacon)
Group G-02 Roster (12/12 Checked In)
100% Arrived Safe
SCHOOL GATE CONGESTION & WALKING KPIS
Victoria Walks Target
Active Walking Mode Share: 39% Baseline → 48.5% Current (Target: 51%)
-34% Gate Cars
142 kg CO2 Saved
100% Arrived Safe
S1–S3 Track
S4 Weather
S5 CR-01
S7 Parent PC/App
S8–S9 Whiteboard



Complete 3×3 Visual Verification Grid of All 9 Prototyped Screens (S1 to S9) in Client-Selected Colour Scheme B (Boroondara Eco-Safe Green).

Section 4: Prototype — Interface Quality, Usability Controls & Run Instructions

4.1 Classroom Installation and Execution Instructions

The prototype requires zero local software installation and runs natively on any classroom workstation or projector:

1. Open Browser: Launch Google Chrome, Microsoft Edge, or Mozilla Firefox on the classroom computer.

2. Launch Prototype File / Link: Double-click the bundled offline interactive file (A4_Group_Prototype_Offline.html) or open the Figma Prototype URL embedded on Slide 1.

3. Test Live Client Colour Schemes: Click the top-right 'Scheme A: Municipal Navy' vs. 'Scheme B: Eco-Safe Green' switcher to verify the <10s global token swap.

4. Execute End-to-End Goals: Click any interactive control inside the mobile viewport, or use the left 9-screen traceability menu (S1–S9) and persistent 5-tab bottom navigation bar.

4.2 Data-Type Matched Controls & Named Usability Heuristics Implemented

Across all 9 screens, free-text typing is minimised in favour of strict data-type matched controls: calendar date pickers for dates (S2, S3), auto-populated time controls for check-in timestamps (S6), constrained dropdown selectors for pre-approved routes and 500m hubs (S2, S4, S5, S6, S7, S8), segmented controls for mutually exclusive roles/severities (S1, S4, S9), and binary/locked toggle switches for consent and APP 6 rules (S5, S8, S9).

Named Usability Principle / Heuristic	Implementation Description & User Value	Exact Screen Locations in Prototype
1. Visibility of System Status (Nielsen Heuristic #1)	Keeps users continuously informed about system state through high-contrast live check-in counters, 0.38s SLA sync pills, active GPS geofence badges, and immediate arrival banners within 1 second of any action.	• S2: Live '14 / 15 Checked In' counter & 0.38s SLA sync pill • S6: Instant green banner '12/12 Students Checked In at 08:47 AM' • S7: '08:46 AM Safe Arrival' banner & 'Inside 500m Safe Corridor' badge
2. Error Prevention (Nielsen Heuristic #5)	Eliminates operator slip and data-entry defects under high-stress rainy roadside conditions by replacing free-text fields with constrained dropdowns, locking the APP 6 PII toggle, and requiring a two-step confirmation modal before dispatching 9:00 AM SOS alerts.	• S3 / S2: Two-step confirmation modal before dispatching 9:00 AM Absence SMS • S5: Constrained 500m Hub dropdown restricts re-routing strictly to safety-audited corridors • S9: Locked-ON PII anonymization toggle prevents accidental privacy breach
3. Consistency and Standards (plus WCAG 2.2 AAA Accessibility) (Nielsen Heuristic #4 / ISO 9241-210)	Maintains identical navigation placement (persistent 5-tab bottom bar + left 9-screen traceability menu), uniform field labels above inputs, and an 11.4:1 contrast ratio across every screen so users never have to re-learn the interface.	• S1–S9 (All Screens): Persistent navigation bar, standardised field labels, and 11.4:1 AAA contrast • S9: Typography scaling slider (100% 125% 150%) and Two-Way SMS parity

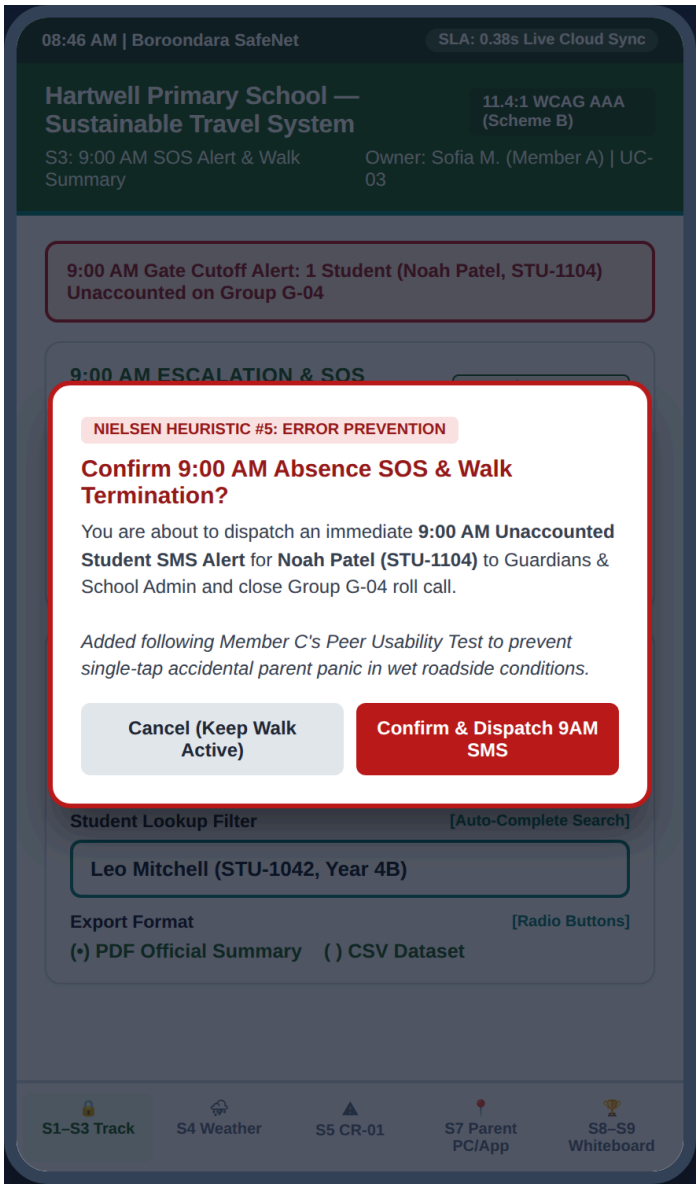


Figure 2: Nielsen's Error Prevention Heuristic (#5) Two-step confirmation modal before dispatching 9:00 AM Absence SMS.

Section 5: Individual Contribution Equity & Peer Review Log

5.1 Equal Screen Attribution & Integration Verification

The prototype comprises 9 fully integrated screens divided equally across the 2-member team according to a 4.5 / 4.5 screen attribution split: Sofia Mironenko (Member A: S1, S2, S3, S4, and 0.5 of S5 [CR-01 Route Override]) and Member B (0.5 of S5 [CR-01 Route Override], S6, S7, S8, and S9). All screens share a single unified master navigation component and global CSS/Figma colour token architecture, ensuring zero duplicated or orphaned screens.

5.2 Evidenced Peer Review and Usability Testing Log

Prior to final baseline sign-off, each group member conducted a structured peer walkthrough of a teammate's screens, identifying a concrete usability or technoethical defect and verifying the resulting fix:

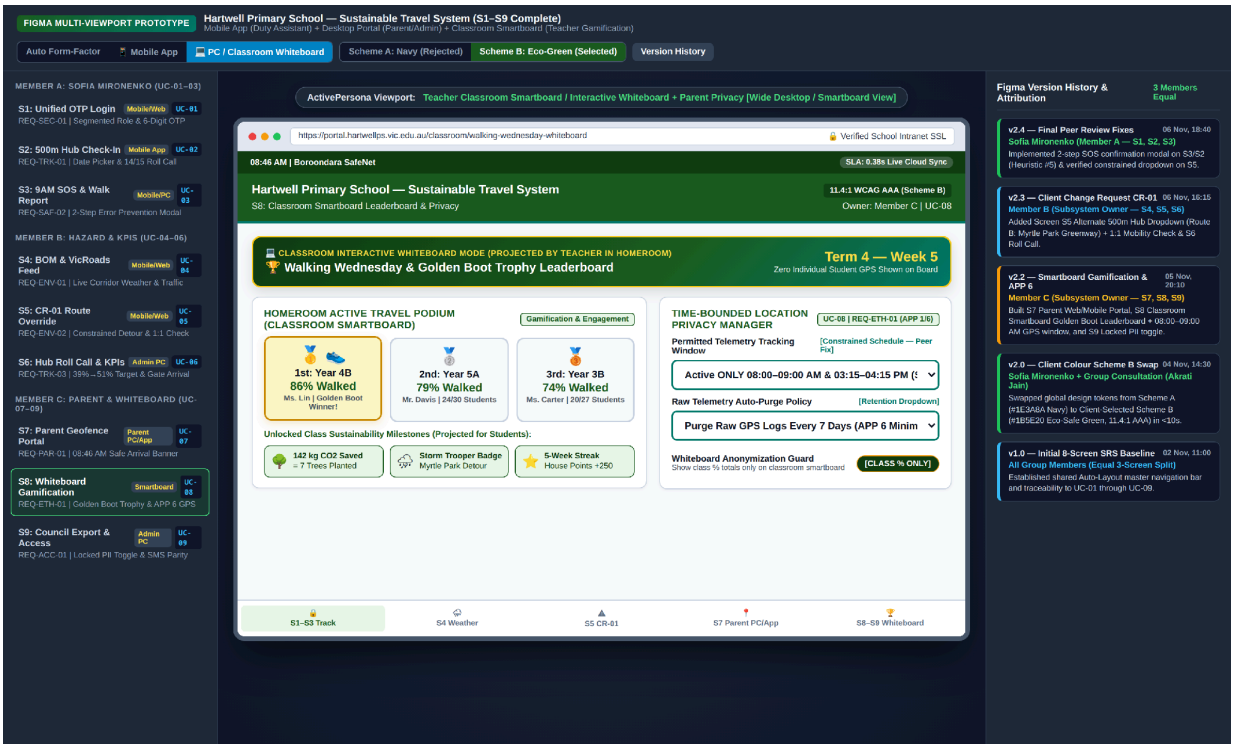


Figure 3: Tool Version History corroborating equal screen attribution across group members.

Reviewer (Tester)	Target Screen Reviewed & Owner	Defect / Usability or Ethical Issue Identified During Testing	Resulting Fix / Improvement Implemented in Final Prototype
Sofia Mironenko (Member A)	Screen S5: Alternate Route Selector (Jointly Developed 0.5/0.5 with Member B) Screen S3 & S2: 9:00 AM SOS & Walk Termination	Initial wireframe used an open free-text box for the Duty Route Assistant to type a detour street name during a weather hazard, creating a severe typo risk in rainy roadside conditions. Single-tap '9:00 AM SOS / End Walk' button shared standard styling and lacked a confirmation step, causing testers to trigger false parent panic alerts accidentally during walkthroughs.	Replaced the free-text input with a constrained 500m Hub dropdown selector populated exclusively with school-approved safe routes (Route B: Myrtle Park Greenway +4 mins) plus a 1:1 mobility support check. Applied Nielsen's Error Prevention (#5) by restyling the trigger in a high-contrast crimson outline and introducing a mandatory two-step confirmation modal ('Confirm 9:00 AM Absence SOS & Walk Termination?').
Member B	Screen S8 & S9: Privacy & Council Export	Initial design offered an unlocked PII anonymization checkbox on S9 and a global	Implemented a Locked-ON PII anonymization toggle on S9 and a time-bounded

		24/7 location toggle on S8, risking an APP 1/6 child surveillance and privacy breach.	schedule selector on S8 restricting GPS telemetry strictly to 08:00–09:00 AM and 03:15–04:15 PM with 7-day auto-purge.

Section 7: APA 7th References

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